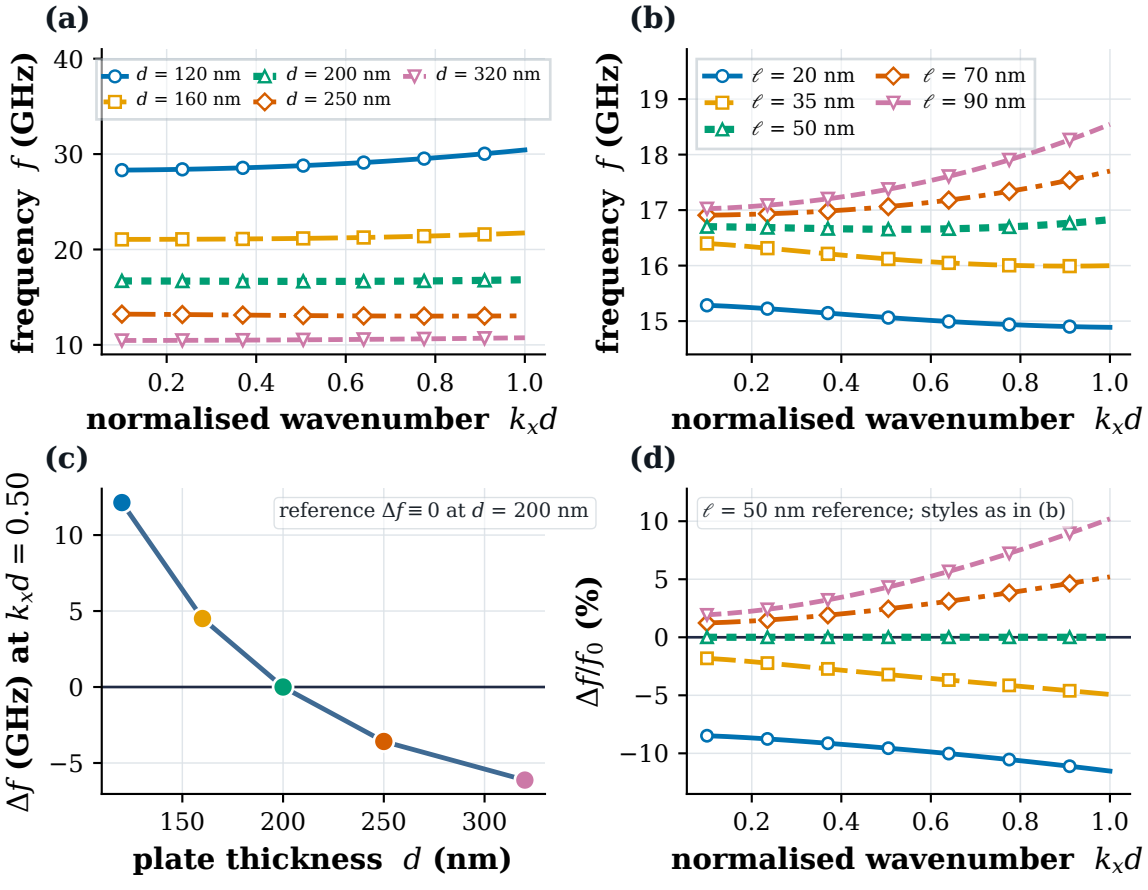
(a)
frequency $f$ (GHz)
40
30
20
10
$d$ = 120 nm
$d$ = 160 nm
$d$ = 200 nm
$d$ = 250 nm
$d$ = 320 nm
0.2
0.4
0.6
0.8
1.0
normalised wavenumber $k_x d$
(b)
frequency $f$ (GHz)
19
18
17
16
15
$\ell$ = 20 nm
$\ell$ = 35 nm
$\ell$ = 50 nm
$\ell$ = 70 nm
$\ell$ = 90 nm
0.2
0.4
0.6
0.8
1.0
normalised wavenumber $k_x d$
(c)
$\Delta f$ (GHz) at $k_x d = 0.50$
10
5
0
−5
reference $\Delta f \equiv 0$ at $d$ = 200 nm
150
200
250
300
plate thickness $d$ (nm)
(d)
$\Delta f / f_0$ (%)
10
5
0
−5
−10
$\ell$ = 50 nm reference; styles as in (b)
0.2
0.4
0.6
0.8
1.0
normalised wavenumber $k_x d$